CS4268

**NATIONAL UNIVERSITY OF SINGAPORE**

**CS4268 – QUANTUM COMPUTING**

(Semester 2: AY2018/19)

Time Allowed : 2 Hours

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**INSTRUCTIONS TO STUDENTS**

1. Please write your Student Number only. Do not write your name.
2. This assessment paper contains **FIVE** questions for a total of **40** marks, and comprises **TWO** printed pages.
3. Students are required to answer **ALL** questions.
4. Students should write the answers for each question on a new page.
5. This is an OPEN BOOK assessment.
6. Permitted devices: electronic calculators

**Question 1. (3+3+3+3 marks)**

For $\theta \in [0, 2\pi)$, let $U_\theta$ denote the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

and let $|\theta\rangle$ and $|\theta^\perp\rangle$ denote $U_\theta|0\rangle$ and $U_\theta|1\rangle$ respectively.

(a) Show that $\sigma_z\sigma_x|\theta^\perp\rangle = |\theta\rangle$, where $\sigma_z$ and $\sigma_x$ are the Pauli matrices:

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(b) Show that the Bell state $|\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ can also be written as $\frac{1}{\sqrt{2}}(|\theta\rangle|\theta^\perp\rangle - |\theta^\perp\rangle|\theta\rangle)$.

(c) Suppose Alice and Bob share the Bell state $|\psi^-\rangle$, and Alice applies $U_\theta^{-1}$ to her half of the Bell state and then measures. What is the state on Bob's side conditioned on her measurement outcomes being 0, and what is his state if the outcome is 1?

(d) Suppose Alice knows the number $\theta$ and Alice and Bob want Bob to end up with the state $|\theta\rangle$. Give a protocol between Alice and Bob for this scenario, using the Bell state $|\psi^-\rangle$ and a *single* classical bit of communication from Alice to Bob, so that Bob has $|\theta\rangle$ at the end of the protocol. Note that we do not care about the overall phase of the state.

**Question 2. (7 marks)**

Give a quantum circuit that reversibly performs the transformation

$$|x_1x_2x_3\rangle|b\rangle \to \begin{cases} |x_1x_2x_3\rangle|1-b\rangle & \text{if } x_1x_2x_3 = 110 \\ |x_1x_2x_3\rangle|b\rangle & \text{otherwise,} \end{cases}$$

for $x_1, x_2, x_3, b \in \{0,1\}$. You are allowed to use every elementary gate you have encountered, such as $\sigma_x, \sigma_y, \sigma_z, \mathrm{CNOT}$ and Toffoli gates. If you use ancilla qubits, they must start in the $|0\rangle$ state and return to the $|0\rangle$ state at the end of the circuit.

**Question 3. (7 marks)**

For a function $f : \{0,1\}^n \to \{0,1\}$, $B_f$ and $Z_f$ are the oracles given by

$$B_f|x\rangle|a\rangle = |x\rangle|a \oplus f(x)\rangle \qquad Z_f|x\rangle = (-1)^{f(x)}|x\rangle$$

where $x \in \{0,1\}^n$ and $a \in \{0,1\}$. Give a circuit to implement $B_f$ using a controlled version of $Z_f$:

$$\text{C-}Z_f|x\rangle|a\rangle = \begin{cases} |x\rangle|a\rangle & \text{if } a = 0 \\ (-1)^{f(x)}|x\rangle|a\rangle & \text{if } a = 1. \end{cases}$$

**Question 4. (7 marks)**

Suppose for $f : \{0,1\}^n \to \{0,1\}$ there are $2^{n-2}$ inputs such that $f(x) = 1$. How many queries to $B_f$ does Grover's algorithm need in order to find such an $x$ with certainty? You don't have to describe Grover's algorithm or give a circuit. Justify your answer using properties of Grover's algorithm that you know.

**Question 5. (7 marks)**

Suppose the function $f : \{0,1\}^n \to \{0,1\}$ satisfies the promise that either (i) $f$ is 0 on the first $2^{n-1}$ inputs (in lexicographic order, i.e., inputs $x$ with the most significant bit $x_1$ being 0) and 1 on the last $2^{n-1}$ inputs, or (ii) the number of 1s in the first $2^{n-1}$ inputs plus the number of 0s in the last $2^{n-1}$ inputs is $2^{n-1}$. Modify the Deustch-Jozsa algorithm to distinguish the cases (i) and (ii) with certainty. You don't need to draw a circuit, a clear description in words is sufficient.

(Hint: Consider a function $\tilde{f}$ related to $f$ which satisfies the conditions for the Deustch-Jozsa algorithm, and try to construct an oracle $B_{\tilde{f}}$ for $\tilde{f}$ using oracle $B_f$ for $f$.)

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**END OF PAPER**